Detecting Conflict-of-Interest Exposure in Industry Payments to Clinicians: A Batch, Streaming, and Machine Learning Pipeline for Mephisto Health System

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Abstract

Mephisto Health System manages clinician conflict of interest with an annual self-attestation form. Staff report their own outside income once a year, on the honor system. The process looks backward, cannot be checked against anything, and produces no ranking, so a compliance office able to review a few hundred people per year has no defensible way to choose which few hundred. This project replaces that guesswork with evidence, using the public national record of what industry actually paid.

The Centers for Medicare & Medicaid Services (CMS) Open Payments general payments file for program year 2025; 16,095,794 payment records covering 1,022,513 clinicians and $2.89 billion in transfers of value was trimmed locally with DuckDB, landed in Amazon S3, and processed in Databricks through a bronze, silver, and gold medallion architecture built on Apache Spark and Delta Lake. Four workloads run from the same curated gold table. A batch layer computes three governance indicators. A Structured Streaming job uses event-time windows and a watermark to raise an alert whenever one manufacturer pays one clinician more than $10,000 inside a 90-day window, producing 12,603 flags worth $445.6 million. An unsupervised Spark MLlib K-means model (k = 4, silhouette = 0.677) scores every clinician on how far they sit from their own specialty's normal behavior and returns a ranked 500-name shortlist. Seventy-two percent of that shortlist was independently flagged by the streaming rules with two methods that share no logic, arriving at largely the same names. Finally, the trained model was deployed as a reusable scoring function and applied to Mephisto's own 24-clinician roster, and its logic was exposed as an interactive risk calculator inside a four-page Microsoft Power BI dashboard.

The system turns more than one million clinicians into a defensible list of 500, a reduction of roughly 2,045 to 1 and is explicitly a triage instrument, not a finding of wrongdoing.

Keywords: conflict of interest, Open Payments, Sunshine Act, medallion architecture, Apache Spark, Structured Streaming, K-means, healthcare compliance

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1. Introduction: Business and Data Domain

1.1 Why Payments from Industry to Doctors Matter

Medical device and drug companies pay clinicians every day, and most of those payments are entirely legitimate. A surgeon is paid to teach other surgeons how to implant a new device. A cardiologist is paid to sit on an advisory board. A researcher receives a grant. None of this is unusual, and none of it is illegal.

The concern is not that money moves. It is what happens when a large amount of money moves from **one** company to **one** clinician, repeatedly, over a short period. At that point a reasonable person may start to wonder whether the clinician still chooses that company's product purely on clinical grounds. The clinician may be entirely honest and usually is. Large-scale studies have nonetheless found that industry payments track prescribing behavior at the population level (DeJong et al., 2016; Ornstein et al., 2016), which is why the topic has been treated as a governance problem rather than an individual one (Institute of Medicine, 2009). That is precisely why the situation is called a conflict of *interest* rather than an act of *wrongdoing*, it describes an appearance and a risk, not a proven behavior.

Three pieces of United States law sit behind this. The *Physician Payments Sunshine Act,* enacted as Section 6002 of the Affordable Care Act (2010), requires manufacturers to report almost every payment they make to physicians and teaching hospitals; CMS publishes the results each year as the Open Payments database. The *Anti-Kickback Statute* (42 U.S.C. § 1320a-7b(b), 2018) makes it a criminal offence to offer or receive anything of value in return for referrals or purchases paid for by federal health care programs. The *Stark Law* (42 U.S.C. § 1395nn, 2018) restricts physician referrals to entities in which the physician has a financial interest. A health system that fails to notice a serious conflict inside its own workforce is exposed on all three fronts, and also to the reputational damage that follows a public story.

1.2 The Organization: Mephisto Health System

Mephisto Health System is a multi-site nonprofit health system employing and credentialing several thousand clinicians. Like most health systems, Mephisto has a compliance office responsible for making sure that the organization's clinicians can be trusted to choose devices, drugs, and treatment plans on clinical merit alone.

Today that office runs on annual self-attestation. Once a year every clinician fills in a form declaring outside financial relationships. The form is signed and filed. Three weaknesses follow directly from that design:

- It looks backward**.** A relationship that begins in February is not visible until the following year's form, by which point purchasing decisions have already been made.
- It cannot be verified**.** The organization has no independent source to check the form against, so an omission that is deliberate or simply forgotten looks exactly like a clean declaration.

- It does not rank**.** Even when every form is returned honestly, the result is a filing cabinet, not a priority list. The compliance office still has no way to say who should be reviewed first.

1.3 The Business Problem

The business question this project answers is deliberately narrow and practical:

> ***Of every clinician in the country, which few hundred should Mephisto's compliance office review first, and can that answer be defended with evidence rather than intuition?***

The constraint that makes this hard is capacity. A compliance review is not a database query; it is interviews, contract requests, and legal time. An office with a handful of reviewers can complete a few hundred reviews in a year. The national population of paid clinicians is over one million. The organization therefore does not need a system that detects everything but rather it needs a system that ranks, so that the limited review capacity lands on the highest-exposure cases first, and so that the choice can be explained to a regulator, a board, or the clinician themselves.

1.4 Why This Is a Big Data Problem

This work could not be done in a spreadsheet, and not only because of size. Four properties push it into big data territory.

- *Volume*. The raw source file is roughly 8 GB of CSV containing 16,095,794 payment records across about 90 columns. Microsoft Excel stops at 1,048,576 rows, so the file cannot even be opened, let alone aggregated.
- *Velocity*. CMS publishes in annual batches, but the *business* question is continuous: a concentration of payments that forms in March should not wait until the following year to be noticed. The system therefore has to be able to process payments as a stream, not only as a yearly report.
- *Variety*. Payment records, clinician identity, specialty taxonomy, manufacturer names, and Mephisto's own internal roster are separate sources that have to be brought into one shape before any of them can be compared.
- *Veracity*. Free-text fields carry inconsistent spellings and casing, specialty is packed into a single slash-delimited string, and identity is not guaranteed unique. None of this can be trusted until it has been cleaned deliberately and the cleaning has been verified.

A distributed engine is the natural answer. Apache Spark splits the file across many machines, does the work in parallel, and returns aggregates in seconds rather than hours.

1.5 The Data Source

The project uses the CMS Open Payments general payments file for program year 2025, downloaded from the CMS open data portal (Centers for Medicare & Medicaid Services, 2025a). Field definitions follow the published CMS data dictionary (Centers for Medicare & Medicaid

Services, 2025b). General payments are transfers of value not tied to research or ownership: consulting fees, speaking fees, travel and lodging, meals, gifts, honoraria, education, and royalties. The file is public, national coverage, and legally mandated. Manufacturers face penalties for failing to report, which makes it a far stronger evidence base than a form a clinician fills in about themselves.

Table 1
Source dataset at a glance

Property	Value
Publisher	Centers for Medicare & Medicaid Services (CMS), Open Payments
Program year	2025 (PY2025), general payments file
Raw size / shape	≈ 8 GB CSV, 16,095,794 rows, ~90 columns
Columns retained	16 (identity, specialty, manufacturer, amount, date, nature of payment)
Distinct clinicians	1,022,513
Total value of payments	$2,894,883,382.23
Legal basis	Physician Payments Sunshine Act, ACA §6002 (42 U.S.C. §1320a-7h)
Licence	Public domain, U.S. Government open data

Note. Row and clinician counts are taken directly from the executed notebook cells reproduced in Appendix C.

One point about the data deserves emphasis up front, because it shapes every design decision that follows. Open Payments records what was paid. It does not record what was prescribed, implanted, or ordered. This dataset can therefore establish *exposure*: a financial relationship of a size and shape that warrants a look and it can never, by itself, establish influence. That distinction is carried through the whole system: every output of this project is a queue for human review.

2. The Proposed Solution

2.1 What the System Does, in Plain Language

The system reads the national payment record, reshapes it so that each row describes one clinician rather than one payment, and then asks four different questions of that same table. Each question corresponds to one workload, and each workload produces something the compliance office can act on.

Table 2
The four workloads and what each one answers

Workload	Question it answers	Output
Batch	How big is the overall exposure, and how concentrated is it?	Three governance indicators for the dashboard
Streaming	Is a single company paying one clinician heavily right now?	12,603 near-real-time concentration flags
Machine learning	Who behaves unlike their own specialty peers?	Ranked 500-name investigation shortlist
Deployment	What is the risk tier of a clinician we care about?	Scored Mephisto roster and a live risk calculator

The four are deliberately different in kind. The batch layer measures. The streaming layer applies a fixed rule that a lawyer can read out loud. The machine learning layer finds shapes no one wrote a rule for. The deployment layer takes the model out of the notebook and puts it in front of the people who have to make the decision. Because the rule-based and the model-based layers share no logic, agreement between them is genuine corroboration rather than the same assumption counted twice.

2.2 Streaming Use Case: Catching Concentration as It Forms

The streaming layer exists because timing matters. A relationship that becomes concerning in March is worth knowing about in March.

The rule is intentionally simple: ***if one manufacturer pays one clinician $10,000 or more inside a 90-day window, raise a flag.*** The $10,000 threshold sits well above ordinary meals and travel and comfortably inside the range of a real consulting or speaking arrangement. The windows are fixed, non-overlapping 90-day periods, and the window is what turns a total into a *rate*. $12,000 spread over three years is a stable relationship, while $12,000 inside one quarter is an intensifying one.

Two Spark Structured Streaming features do the real work. ***Event-time windowing*** groups payments by the date the payment actually happened rather than the moment the record arrived, so a late file does not distort the picture. A ***watermark*** of 30 days tells Spark how long to keep a window open for stragglers before closing it and releasing the memory; without a watermark, a stream that runs indefinitely would hold every window it had ever seen. The job writes each flag to a Delta table with a checkpoint, so it can be stopped and restarted without double-counting.

Because the CMS file is a historical archive rather than a live feed, the silver table was rewritten as 24 separate files and replayed one file per trigger. Spark's streaming engine cannot tell the difference between that and a real feed, so the code is production-shaped even though the source is static.

2.3 Batch Use Case: Three Governance Indicators

The batch layer answers the questions a compliance committee asks at the start of a meeting. Each indicator is one Spark SQL aggregation over the gold table.

- KPI 1: Total industry exposure. The full value of general payments in the program year. This is the denominator for everything else and the number that goes on the first line of a board report.
- KPI 2: Share of clinicians above materiality. The proportion of clinicians who received more than $5,000 in total. Below that level a payment record is usually meals and conference travel; above it, a relationship exists that a compliance office would want to know about. The threshold makes the population reviewable.
- KPI 3: Share above their own specialty average. The proportion of clinicians paid more than the mean for their own specialty. This is the indicator that makes the analysis fair: $30,000 is unremarkable for an orthopedic surgeon working with implant manufacturers and highly unusual for a pediatrician. Comparing every clinician only to peers in the same specialty removes that distortion.

KPI 3 is also the conceptual bridge to the machine learning layer, because peer-relative rank is exactly the feature the model clusters on.

2.4 Machine Learning Use Case: Ranking Peer Deviation

Rules only catch what someone thought to write down. The machine learning layer exists to find the shapes nobody specified in advance.

The approach is unsupervised, and that choice is forced by the data rather than preferred. Supervised learning needs labelled examples: a column marking which clinicians were later found to have a genuine conflict. No such column exists, and no health system publishes one. So the model is not asked *who is guilty*; it is asked who does not look like everybody else.

Spark MLlib's **K-means** algorithm (Lloyd, 1982; Meng et al., 2016) groups clinicians into k clusters by similarity on two peer-relative features: percentile rank within their own specialty, and the share of their money that came from a single manufacturer. Values of k from 2 to 8 were tested and scored with the silhouette measure (Rousseeuw, 1987), which rewards clusters that are tight internally and well separated from each other. Four clusters scored best.

The cluster containing the lowest combined risk is then identified programmatically as the baseline: the shape of a normal clinician, and every clinician in the country is scored by Euclidean distance from that baseline center. The 500 highest scores become the investigation

shortlist. Ranking, rather than a yes/no label, is what makes the output usable by a team with fixed capacity.

2.5 Deployment Use Case: The Risk Calculator

A model that only runs inside a notebook helps the person who wrote the notebook. The deployment stage turns the trained model into something a compliance analyst can use without opening Databricks at all.

Three things happen in this stage. First, the specialty-level distributions computed during feature engineering: the peer averages and percentile boundaries for the 57 specialties needed to score the roster are cached so that any new clinician can be positioned against their own peer group instantly. Second, a single reusable function, `classify_clinician()`, takes a specialty, an annual payment total, and the share coming from the largest single company, converts them into the same two peer-relative features the model was trained on, assigns the nearest cluster, and returns a risk tier of HIGH, MEDIUM, or LOW together with the reasoning behind it. Third, that function is wired to interactive controls so a user can type in any combination and read the answer immediately.

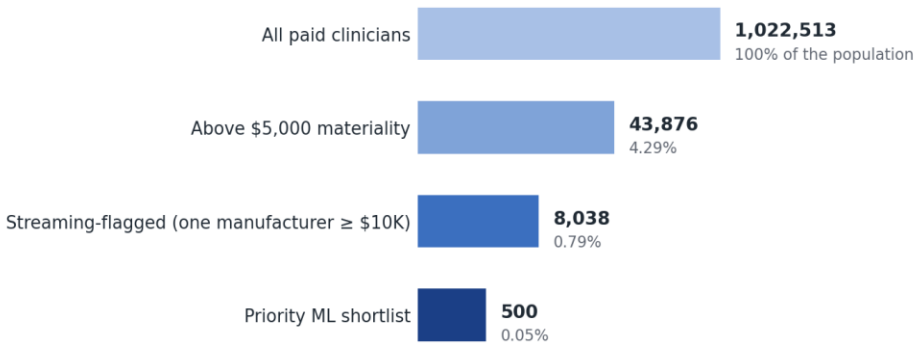
The function was validated against the trained Spark model rather than trusted on assertion. For each test case the cluster assigned by the lightweight scoring logic was compared with the cluster the fitted K-means model returns for the same feature vector, and the notebook prints a match confirmation. The same function was then applied in bulk to Mephisto's own 24-clinician roster, which is stored as a CSV in the bronze layer; all 24 were scored, none were skipped, and the results were written back to gold as `mephisto_scored` for the dashboard to read.

This is the step that changes the character of the deliverable. Without it the project produces a report about the national population. With it, the compliance office can point at one of its own clinicians, or at a hypothetical arrangement being negotiated right now, and get a tier back.

2.6 How the Four Workloads Fit Together

Seen together, the four layers form a funnel. Each layer measures the full population independently, none of them is fed a filtered list by the layer above, so the funnel is a picture of how selective each method is, and a name that appears at more than one level has been identified on more than one kind of evidence.

Figure 1
Narrowing one million clinicians to a reviewable shortlist



Note. Percentages are of the full population of 1,022,513 paid clinicians. The streaming stage counts distinct clinicians flagged, not individual flags; 12,603 flags map to 8,038 people.

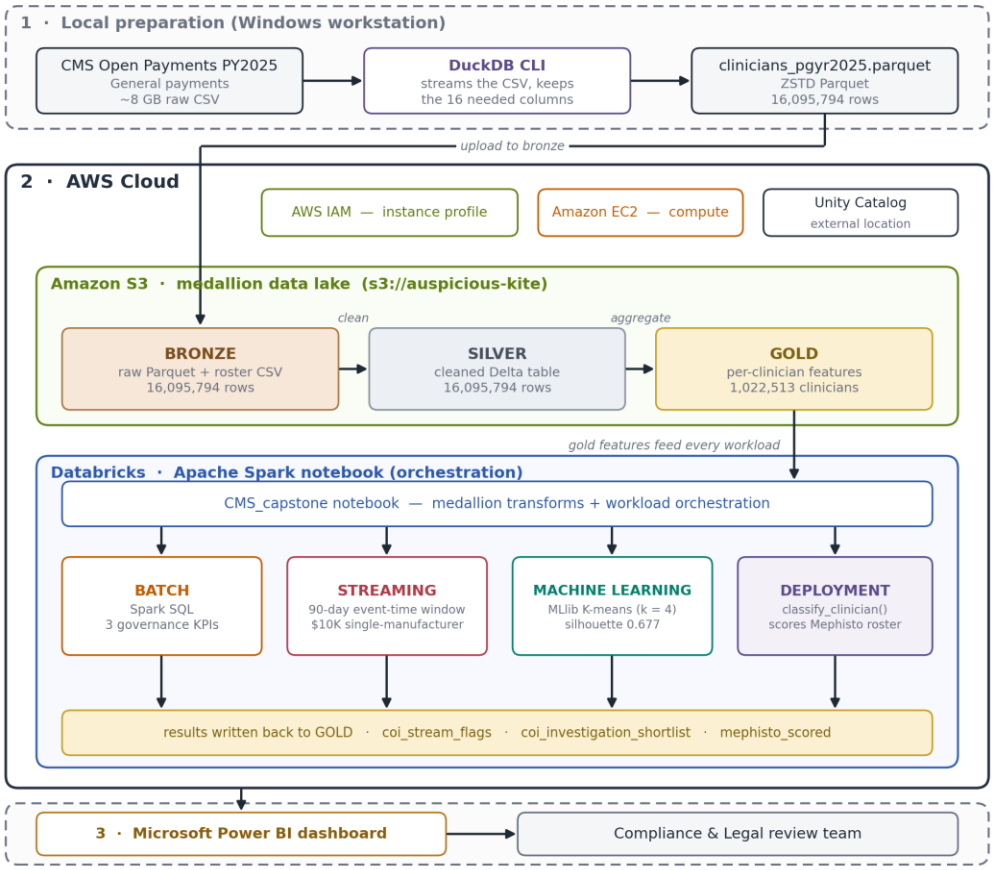
The final shortlist represents 0.05% of the paid population which is a reduction of about 2,045 to 1. That is the number that makes the compliance office's capacity problem tractable.

3. Scope and Design

This section describes the architecture: where the data goes, what each component does, and why each component was chosen over the alternatives.

3.1 The Data's Journey, End to End

Figure 2
End-to-end architecture from raw CMS file to compliance review



Note. Solid boxes are components; dashed enclosures are environments. The gold feature table is the single source that all four workloads read, and three of the four write their results back to gold.

The journey has five movements.

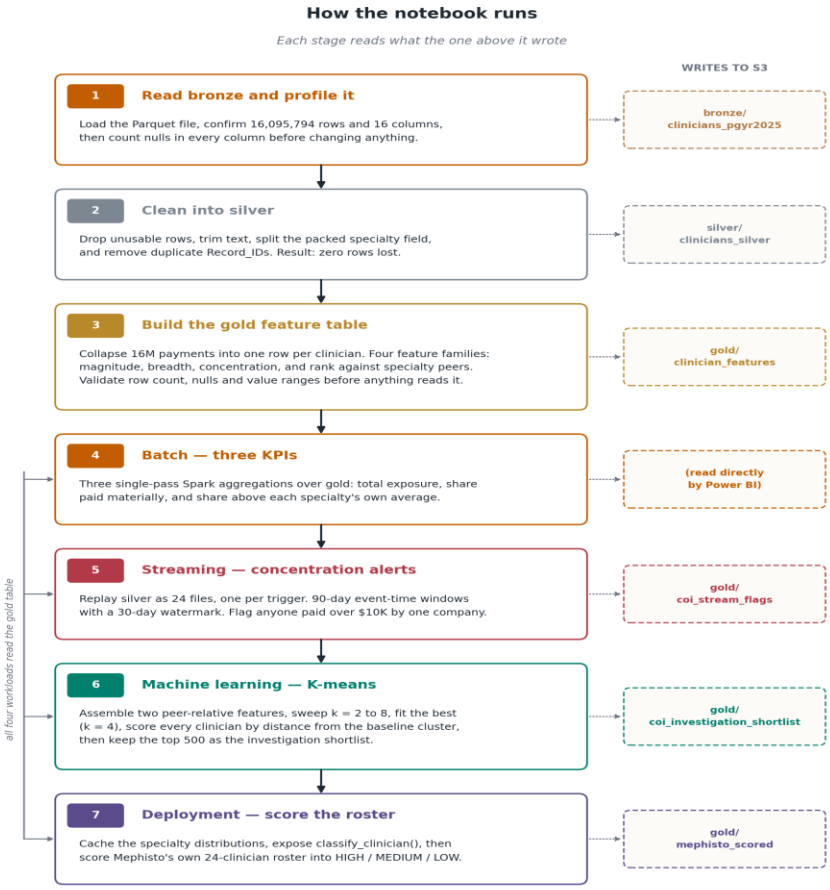
1. *Local preparation*. The raw CSV is streamed through the DuckDB command-line tool on a Windows workstation, which keeps only the 16 columns the project needs and writes compressed Parquet.
2. *Landing.* The Parquet file is uploaded to the bronze prefix of an Amazon S3 bucket, `s3://auspicious-kite`, alongside Mephisto's own roster CSV. Bronze is never modified after this point.
3. *Refinement*. A Databricks notebook running Apache Spark reads bronze, cleans it into a silver Delta table, and aggregates silver into a gold table with one row per clinician.
4. *Analysis*. Batch, streaming, machine learning, and deployment workloads all read gold. Streaming, machine learning, and deployment write their results back into gold as new tables.

5. *Consumption*. Microsoft Power BI reads the gold tables and presents them to the compliance and legal review team across four pages.

3.2 How the Notebook Runs

The architecture diagram shows *where* the data lives. The diagram below shows *what the code does*, stage by stage, in the order the notebook executes. Each stage reads what the stage above it wrote, which is why a failure at any point stops everything downstream rather than silently producing partial results.

Figure 3
Execution flow of the Databricks notebook



Note. Stages 1 to 3 build the medallion layers. Stages 4 to 7 are the four analytical workloads, all reading the same gold feature table.

3.3 What Each Component Does, and Why It Was Chosen

DuckDB: local pre-processing

Job: read the 8 GB CSV in a streaming fashion, discard about 74 unused columns, and write ZSTD-compressed Parquet.

Why: DuckDB is an in-process analytical engine designed for exactly this kind of local reduction (Raasveldt & Mühleisen, 2019). Uploading 8 GB of CSV to the cloud and then discarding most of it wastes transfer time, storage, and every subsequent read. DuckDB does this on the workstation with no cluster running, so the reduction costs nothing in cloud compute. It also converts row-oriented CSV into column-oriented Parquet, which is the format Spark reads most efficiently — a query touching four columns reads only those four.

Amazon S3: the data lake

Job: hold every version of the data, from raw Parquet through to the final scored tables, in a single bucket organized by medallion layer.

Why: S3 separates storage from compute (Amazon Web Services, 2025). The data persists whether or not a cluster is running, which means the cluster can be shut down between sessions without losing anything which is the single largest cost control in the whole design. S3 is also durable, effectively unlimited, and readable by every tool in the stack.

The medallion architecture: bronze, silver, gold

Job: impose three named quality levels on the lake so that no consumer ever has to guess how trustworthy a table is.

Why: the medallion pattern (Databricks, 2025a) is a naming discipline as much as a design. Bronze holds raw, immutable data exactly as received; if a cleaning rule turns out to be wrong, the original is still there and the pipeline can be rerun. Silver holds cleaned, validated records at the same grain as bronze. Gold holds the aggregated, business-ready feature table. The discipline is what keeps four different workloads consistent with each other. They read one gold table, so they cannot disagree about the underlying facts.

Delta Lake: the table format

Job: add database behavior to files sitting in object storage.

Why: plain Parquet has no notion of a transaction. Delta adds an ACID transaction log (Armbrust et al., 2020), which matters enormously here because the streaming job writes continuously while the dashboard reads. Without it, a reader could catch a half-written file. Delta also provides schema enforcement, which stops a malformed batch from corrupting a table, and time travel, which allows any earlier version of a table to be read back.

Databricks and Apache Spark: the processing engine

Job: run every transformation, aggregation, streaming query, and model fit.

Why Spark: it splits work across many machines and executes in parallel (Zaharia et al., 2016), which is the only practical way to aggregate 16 million rows repeatedly. It also handles batch, streaming, SQL, and machine learning in one engine and one language, so the project does not need four separate systems that must be kept in agreement.

Why Databricks: it supplies the managed cluster, the notebook interface, Delta Lake, MLlib, and Unity Catalog governance in one workspace, and it connects to S3 through an instance profile rather than static keys.

AWS IAM and Unity Catalog: access and governance

Job: control who and what may read or write the bucket.

Why: an IAM instance profile attached to the cluster grants S3 access without any credential ever appearing in the notebook. Unity Catalog registers the bucket as a governed external location, so access is granted centrally and every read is attributable. For healthcare data this is not optional polish; it is the audit trail.

Spark Structured Streaming: the near-real-time layer

Job: consume payments incrementally and emit concentration flags.

Why: Structured Streaming uses the same DataFrame API as the batch code (Databricks, 2025b), so one set of transformations serves both paths. Its event-time and watermark support are what make a 90-day window trustworthy when data arrives out of order, and checkpointing makes restarts safe.

Spark MLlib: the machine learning layer

Job: cluster one million clinicians and score each one against the baseline.

Why: MLlib runs distributed on the same cluster and reads the same DataFrames, so no data has to be exported to a separate machine learning environment. A single-machine library such as scikit-learn would require pulling a million rows into one process, which reintroduces exactly the bottleneck Spark exists to remove.

Microsoft Power BI: the delivery layer

Job: present the results to compliance staff, who are lawyers and analysts rather than engineers.

Why: the analysis is worthless if its audience cannot reach it. Power BI (Microsoft, 2025) provides filtering, drill-down, and export in a familiar interface, and reads the gold Delta tables directly, so the dashboard reflects the same numbers as the pipeline with no manual re-keying.

3.4 Data Wrangling: Bronze to Silver to Gold

Every transformation in this section was verified with an explicit check in the notebook, not assumed.

Bronze: land it and look at it

The Parquet file is read and profiled before anything is changed: row count, column count, data types, and a null count for every column. Confirming 16,095,794 rows and 16 columns at this stage establishes the number every later step is checked against.

Silver: clean it

Five operations turn bronze into silver.

- Drop unusable rows. Records missing a clinician identifier, a payment amount, or a payment date cannot support any analysis in this project and are removed. In this dataset the count removed was zero, which is itself a useful finding about CMS data quality.
- Standardize text. Names, cities, states, and manufacturer names are trimmed of whitespace and cased consistently, so that the same manufacturer does not appear as three different entities in a group-by.
- Split the packed specialty field. CMS stores specialty as a single slash-delimited string such as `Allopathic & Osteopathic Physicians/Orthopaedic Surgery/Adult Reconstructive Orthopaedic Surgery`. This is split into taxonomy level, primary specialty, and subspecialty, which is what makes peer comparison possible at all.
- Remove duplicates. Records are de-duplicated on `Record_ID`. Zero duplicates were found.
- Cast types**.** Amounts become double precision and dates become true date types, so that arithmetic and event-time windowing behave correctly.

Silver is written as a Delta table and its row count re-checked. Because no rows were dropped and no duplicates found, silver holds the same 16,095,794 records as bronze; cleaned, typed, and structured.

Gold: aggregate it into features

Gold collapses 16 million payments into 1,022,513 rows, one per clinician, described by 19 columns. The features fall into four families.

Table 3
Feature families in the gold table

Family	What it captures	Representative features
Magnitude	How much money	total received, number of payments, average and largest payment
Breadth	How many sources	distinct manufacturers, distinct payment natures
Concentration	How dependent on one source	largest single manufacturer total, share of money from that manufacturer
Peer relativity	How they compare to their own specialty	specialty mean, percentile rank within specialty, z-score against the specialty mean

Concentration and peer relativity are the two families that carry the analysis. Magnitude alone is a poor signal: the highest-paid clinicians in the country are often exactly the people you would expect to be highly paid; senior surgeons who genuinely do consult widely. Concentration and peer-relative rank are what separate an unusual pattern from a merely large one.

Gold is validated before anything reads it: row count, null count per column, and range checks confirming that percentiles fall between 0 and 1 and shares between 0 and 1.

4. Results

Every figure in this section is taken from an executed notebook cell. The full outputs are reproduced in Appendix C.

4.1 Pipeline Integrity

Before any analytical result can be trusted, the pipeline itself has to be shown to be sound.

Table 4
Row counts and validation checks at each medallion layer

Layer	Rows	Columns	Validation result
Bronze	16,095,794	16	Profiled; null counts recorded for every column
Silver	16,095,794	18	0 rows dropped, 0 duplicate Record_IDs, types cast
Gold	1,022,513	19	0 nulls in key features; percentiles and shares within 0–1

No records were lost between bronze and silver. That is worth stating plainly, because a cleaning stage that quietly discards rows is one of the most common ways an analysis becomes wrong without anyone noticing. The gold row count of 1,022,513 is the number of distinct clinicians in the file, and it is the denominator for every percentage that follows.

4.2 Batch Results: The Three Indicators

Table 5
Batch governance indicators, CMS Open Payments PY2025

Indicator	Result	Reading
KPI 1 — Total industry exposure	$2,894,883,382.23	The total value of general payments in the program year
KPI 2 — Above $5,000 materiality	43,876 clinicians (4.29%)	Fewer than one in twenty clinicians received a material amount
KPI 3 — Above own specialty average	125,919 clinicians (12.3%)	About one in eight is paid above their own peer group's mean

The first two indicators together tell an encouraging story: the great majority of the $2.89 billion sits in small payments, and only 4.29% of clinicians cross a level that a compliance office would consider material. Concentration, not breadth, is the problem to manage.

The third indicator is the one that changes how the population is read. Twelve percent sit above their own specialty's mean, but that share varies enormously by specialty, and it is the variation that is informative.

Table 6
Selected specialties: peer average and share paid above it

Specialty	Clinicians	Specialty average	% above average
Independent Medical Examiner	11	$24,464.51	9.09%
Orthopaedic Surgery	24,950	$23,498.60	9.10%
Neurological Surgery	5,539	$16,891.15	11.12%
Electrodiagnostic Medicine	12	$8,619.58	25.00%
Internal Medicine	153,962	$4,882.69	8.46%

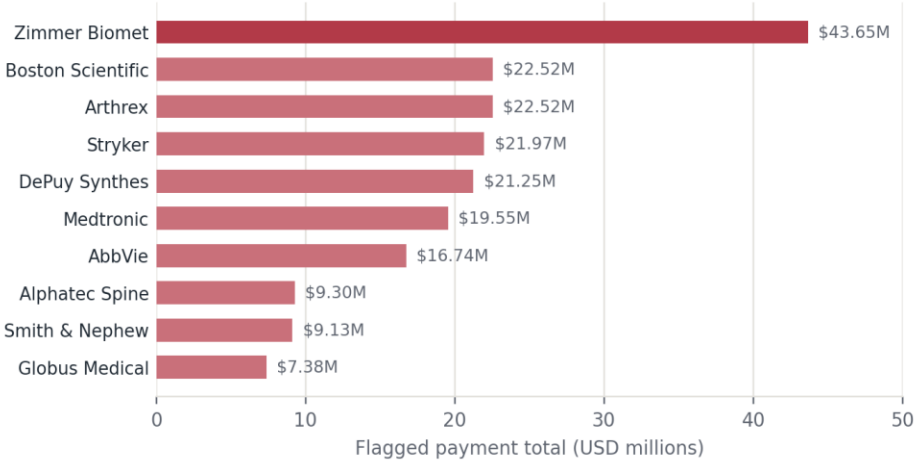
Read the orthopedic and internal medicine rows together. The average orthopedic surgeon receives roughly five times what the average internist receives. A flat dollar threshold applied to both would fill a review queue with orthopedic surgeons behaving completely normally for their field, and would miss an internist receiving an amount that is extraordinary for theirs. This single comparison is the strongest argument in the project for peer-relative measurement, and it is why the machine learning model is built on rank rather than on dollars.

4.3 Streaming Results: Concentration Flags

The streaming job replayed the payment history as 24 incremental batches and produced 12,603 flag events, covering 8,038 distinct clinicians and 131,270 individual payments worth $445,644,622.64. The average flagged 90-day window was approximately $35,000. No records were dropped by the watermark, confirming that the 30-day allowance was generous enough for this data's arrival pattern.

Two readings matter here. First, 8,038 clinicians is 0.79% of the paid population. The rule is selective rather than indiscriminate. Second, $445.6 million is 15.4% of all money in the file concentrated on that 0.79%, which is precisely the shape the rule was designed to expose.

Figure 4
Top ten manufacturers by total flagged payment value



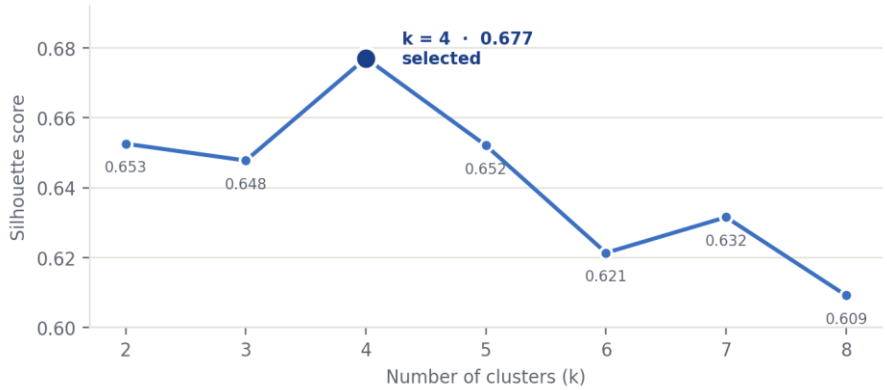
Note. Values are the sum of flagged 90-day window totals attributed to each manufacturer, not the manufacturer's total spend.

The flagged value is dominated by orthopedic and cardiovascular device makers. Zimmer Biomet alone accounts for $43.65 million, roughly twice the next manufacturer. This is a coherent finding rather than a surprising one: implant selection is surgeon-driven and device-specific, which makes the surgeon relationship commercially valuable in a way it is not in most fields. For a compliance office it points to where policy attention is best spent; the device service lines rather than the whole organization.

4.4 Machine Learning Results: Four Behavioral Clusters

Silhouette scores were computed for k from 2 to 8. Four clusters scored highest at **0.677**, which indicates well-separated and internally coherent groups.

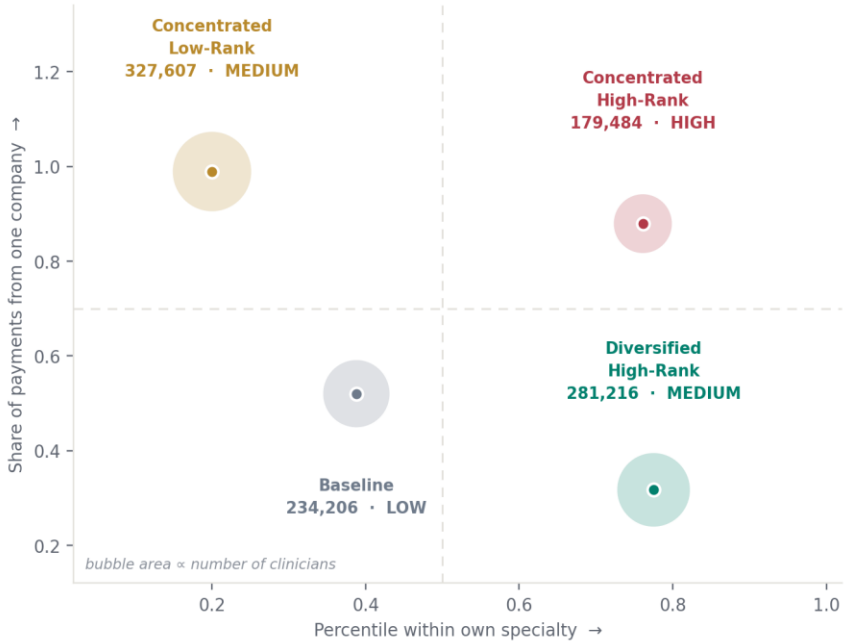
Figure 5
Silhouette score by number of clusters



Note. Higher is better. The score rewards clusters that are compact internally and distant from one another.

The four clusters divide the population along the two axes the model was given: how highly a clinician ranks within their own specialty, and how much of their money comes from a single company.

Figure 6
The four behavioural clusters, positioned by the two model features



Note. Bubble area is proportional to the number of clinicians. Dashed guides mark the 0.50 percentile and 0.70 single manufacturer share boundaries used to name the clusters.

Table 7
Cluster profiles and assigned risk tiers

Cluster	Clinicians	Mean percentile	Mean single-mfr share	Tier
Concentrated High-Rank	179,484	0.761	0.879	HIGH
Diversified High-Rank	281,216	0.775	0.317	MEDIUM
Concentrated Low-Rank	327,607	0.200	0.989	MEDIUM
Baseline	234,206	0.388	0.520	LOW

Each cluster has a plain-English meaning:

- Concentrated High-Rank: paid well above their peers, and most of it from one company. This is the classic conflict-of-interest shape and the only tier rated HIGH.
- Diversified High-Rank: also paid well above their peers, but from many companies. Consulting widely is a legitimate profile, so the tier is MEDIUM rather than HIGH.
- Concentrated Low-Rank: nearly all income from one company, but small amounts. Usually, a single manufacturer's meals or travel. Worth monitoring, not investigating first.

- Baseline: unremarkable on both measures. This is the majority behavior and the reference point the anomaly score is measured against.

The baseline cluster was identified programmatically, by ranking the four centers on the sum of their two feature values and taking the lowest, rather than being chosen by eye. Every clinician was then scored by Euclidean distance from that center. Scores ranged up to 0.7775, and the 500 highest became the investigation shortlist.

Every one of the 500 belongs to the Concentrated High-Rank cluster. That is a good sign rather than a redundant one: the anomaly score and the cluster assignment are computed differently, and their agreement means the ranking is not drifting away from the behavior it was meant to capture.

4.5 Cross-Validation: Two Methods, One Answer

The most important result in the project is not any single number rather it is the overlap between two detectors that share no logic.

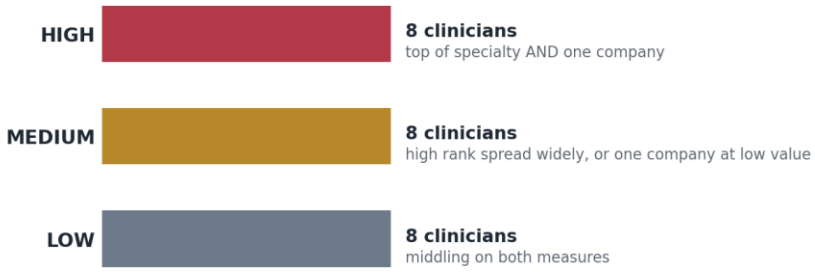
*362 of the 500 shortlisted clinicians (72.4%) were independently flagged by the streaming rules***.** The streaming layer knows nothing about specialty percentiles, clusters, or distances; it applies a fixed dollar threshold inside a time window. The machine learning layer knows nothing about $10,000 or 90 days; it measures peer-relative deviation across a whole year. Agreement at 72% is therefore genuine corroboration.

The remaining 138 names are just as interesting. They are clinicians whose overall annual pattern is highly unusual for their specialty but who never crossed the $10,000 single-window line, steady payments that stay under the rule, which is exactly the population a threshold cannot see and a model can. This is the clearest practical justification for running both methods rather than choosing one.

4.6 Deployment Results: Scoring Mephisto's Own Roster

The deployment stage cached peer distributions for 57 specialties and applied `classify_clinician()` to Mephisto's roster of 24 clinicians, held as a CSV in the bronze layer. All 24 were scored successfully and none were skipped for missing specialty coverage.

Figure 7
Risk tiers assigned to the Mephisto clinician roster



Note. The roster is fictional test data constructed to exercise every branch of the scoring logic; it does not describe real individuals.

The even split across three tiers is by design. The roster was built to contain clear cases of each shape so that every branch of the scoring function is exercised. The value of the exercise is not the distribution but the demonstration: a real roster can be dropped in the same place and scored the same way, against a model trained on the full national population.

The interactive calculator was checked case by case against the trained model. In one worked example a hypothetical denturist receiving $80,000 in a year, all of it from a single manufacturer, is placed at the 100th percentile of their specialty with a single-manufacturer share of 1.00, assigned to Concentrated High-Rank, and returned as HIGH, with the notebook printing a confirmation that the lightweight Python scoring path agrees with the fitted Spark model on that same feature vector.

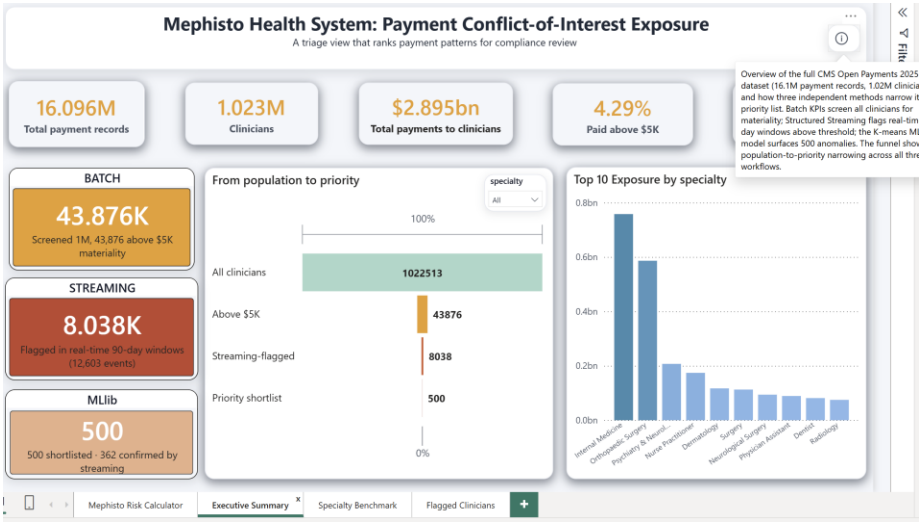
5. The Dashboard (here)

The dashboard is where the pipeline meets its actual users. The people who act on this work are compliance officers and lawyers, not engineers, so every page was built to answer one question and to be readable without training. The four pages are described below in analytical order, from the widest view to the most specific, although the report file opens on the risk calculator.

Two design rules run through all four pages. First, every page carries an information button that states, in one paragraph, what the page shows and where the numbers come from, so a reader who arrives cold is not guessing. Second, every page repeats the same caveat: this is prioritization for review, not proof of wrongdoing.

5.1 Executive Summary: the whole picture on one screen

Figure 8
Executive Summary page



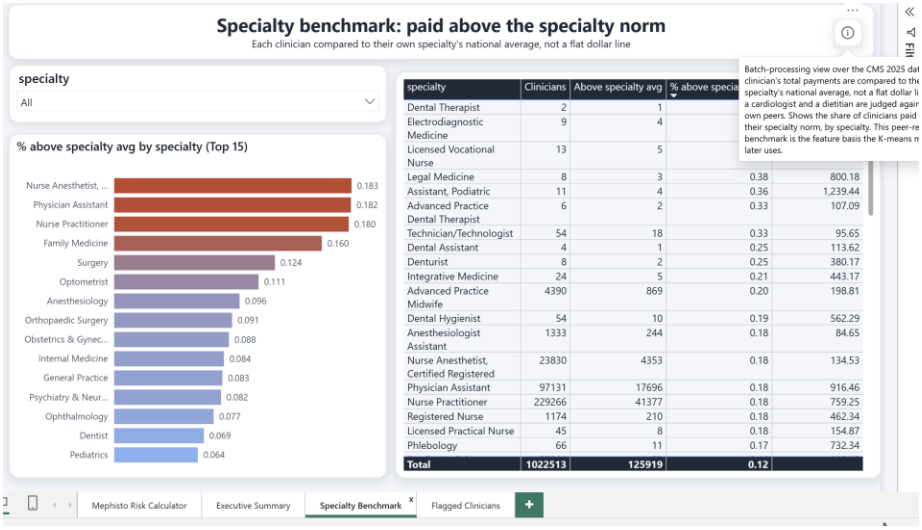
Note. The card strip carries the four headline figures; the left column shows what each of the three national workloads contributed; the centre funnel shows the narrowing from population to priority.

What it shows. The scale of the problem and how far the system narrows it. The top cards give the four numbers a board asks for: 16.096 million payment records, 1.023 million clinicians, $2.895 billion in total payments, and 4.29% paid above the $5,000 materiality line.

How to read it. The left column separates the three national workloads so their contributions can be seen independently. Batch screened one million and returned 43,876 above materiality; streaming raised 12,603 flag events covering 8,038 clinicians; the MLlib model shortlisted 500, of which 362 were confirmed by streaming. The center funnel puts those same stages on one scale so the 2,045-to-1 reduction is visible rather than asserted. The right-hand chart ranks total exposure by specialty, showing that internal medicine and orthopedic surgery carry the largest absolute dollars, which is a volume effect and precisely why the next page reframes everything in peer-relative terms.

5.2 Specialty Benchmark: comparing like with like

Figure 9
Specialty Benchmark page



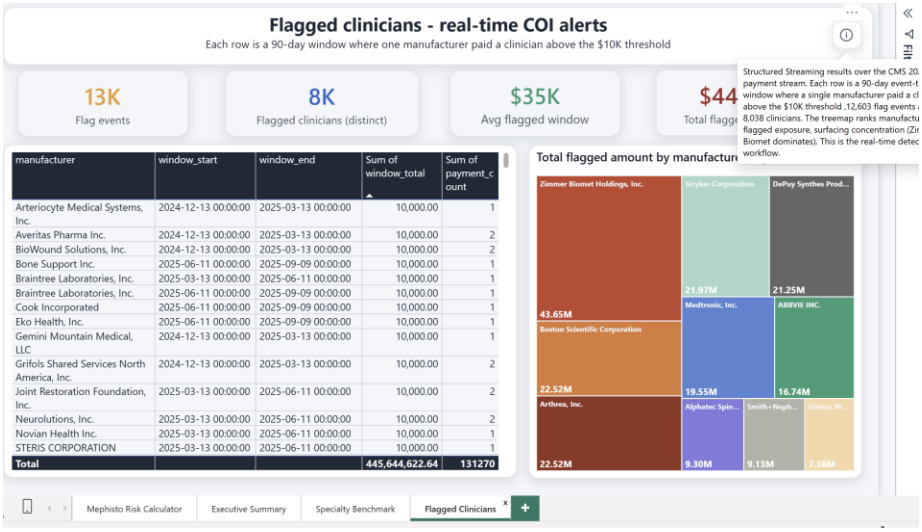
Note. Left: the fifteen specialties with the highest share of clinicians paid above their own norm. Right: the full sortable table with clinician counts, counts above average, share, and the specialty's average payment.

What it shows. Each clinician measured against their own specialty's national average rather than against a flat dollar line.

How to read it. The bar chart answers "which specialties have the widest internal spread?" and produces a genuinely counter-intuitive answer: among the larger specialties, nurse anesthetists, physician assistants, and nurse practitioners top the list at around 18%, ahead of the surgical specialties. That is not because these clinicians are paid more, but because their payment distributions are more uneven. The table on the right holds the underlying numbers for all specialties and can be sorted or filtered, with the total row confirming 125,919 of 1,022,513 clinicians (12.3%, shown rounded to 0.12) above their own specialty average. This page is the visual argument for the feature the machine learning model is built on.

5.3 Flagged Clinicians: the near-real-time alert queue

Figure 10
Flagged Clinicians page



manufacturer	window_start	window_end	Sum of window_total	Sum of payment_count
Arteriocyte Medical Systems, Inc.	2024-12-13 00:00:00	2025-03-13 00:00:00	10,000.00	1
Averitas Pharma Inc.	2024-12-13 00:00:00	2025-03-13 00:00:00	10,000.00	2
BioWound Solutions, Inc.	2024-12-13 00:00:00	2025-03-13 00:00:00	10,000.00	2
Bone Support Inc.	2025-06-11 00:00:00	2025-09-09 00:00:00	10,000.00	1
Braintree Laboratories, Inc.	2025-03-13 00:00:00	2025-06-11 00:00:00	10,000.00	1
Braintree Laboratories, Inc.	2025-06-11 00:00:00	2025-09-09 00:00:00	10,000.00	1
Cook Incorporated	2025-06-11 00:00:00	2025-09-09 00:00:00	10,000.00	1
Eko Health, Inc.	2025-06-11 00:00:00	2025-09-09 00:00:00	10,000.00	1
Gemini Mountain Medical, LLC	2024-12-13 00:00:00	2025-03-13 00:00:00	10,000.00	1
Grifols Shared Services North America, Inc.	2024-12-13 00:00:00	2025-03-13 00:00:00	10,000.00	2
Joint Restoration Foundation, Inc.	2025-03-13 00:00:00	2025-06-11 00:00:00	10,000.00	2
Neurolutions, Inc.	2025-03-13 00:00:00	2025-06-11 00:00:00	10,000.00	2
Novian Health Inc.	2025-03-13 00:00:00	2025-06-11 00:00:00	10,000.00	1
STERIS CORPORATION	2025-03-13 00:00:00	2025-06-11 00:00:00	10,000.00	1
Total			**445,644,622.64**	**131270**

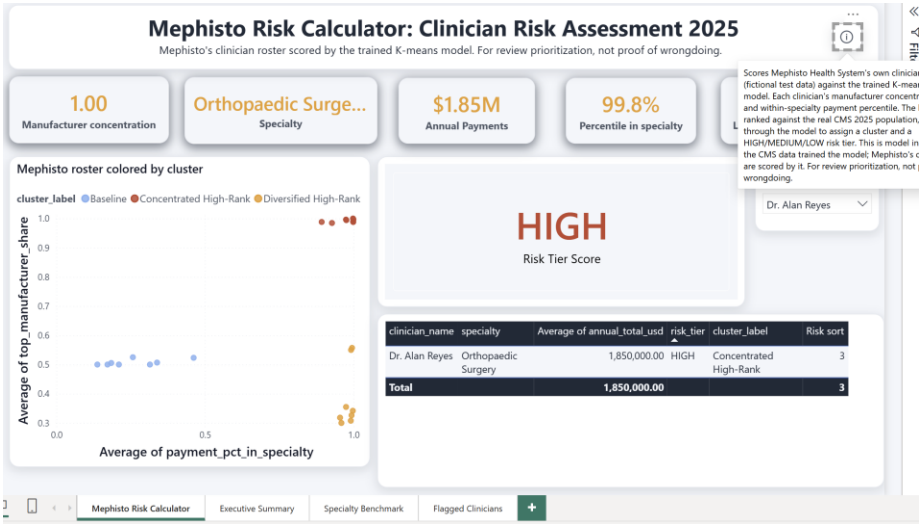
Note. Each table row is one 90-day event-time window in which a single manufacturer paid one clinician above the $10,000 threshold. The treemap sizes manufacturers by total flagged value.

What it shows. The output of the streaming workload of 13 thousand flag events across 8 thousand distinct clinicians, an average flagged window of $35,000, and $445.6 million in total flagged value.

How to read it. The table is the working queue: manufacturer, window start and end, the window total, and how many payments made it up. A reviewer can sort by value to work the largest exposures first or scan the window dates to see whether a relationship is intensifying. The tree map on the right answers the portfolio question which manufacturers account for the most flagged value and shows Zimmer Biomet at $43.65 million dominating a field where the next tier sits between $16 million and $23 million. The total row anchors the page at $445,644,622.64 across 131,270 payments.

5.4 Mephisto Risk Calculator : the deployed model

Figure 11
Mephisto Risk Calculator page



Note. The page shown has an information tooltip open over the right-hand controls. Selecting a clinician updates the card strip, the large risk tier, and the detail table.

What it shows. The trained model applied to Mephisto's own people rather than to the national population. This is the page that turns the project from analysis into a tool.

How to read it. A reviewer selects a clinician from the control on the right. The card strip fills with that person's inputs and derived values: manufacturer concentration, specialty, annual payments, and percentile within specialty and the large panel in the center returns the risk tier in a single word. In the example shown, Dr. Alan Reyes, an orthopedic surgeon with $1.85 million in annual payments, a manufacturer concentration of 1.00, and a 99.8th percentile position within his specialty, is assigned to the Concentrated High-Rank cluster and returned as HIGH. The scatter plot on the left places the whole roster on the same two axes the model uses, colored by cluster, so an individual can be seen in the context of their colleagues.

The roster on this page is fictional test data. Its purpose is to demonstrate that the deployed model accepts an arbitrary clinician list and returns a defensible tier for each name.

6. Recommendations

The recommendations below are ordered by how quickly they can be acted on. Each states what to do, why it follows from the results, and what it costs. They are written to be adopted by a compliance office of the size Mephisto has, not an idealized one.

6.1 Immediate (0–3 months)

R1. Replace self-attestation as the primary triage signal, not as the record

Keep the annual form. It captures relationships Open Payments does not, such as equity in a private company or an unpaid advisory role. But stop using it to decide who gets reviewed. Use the 500-name shortlist and the streaming queue for that and use the form as a cross-check: a clinician who appears in the top tier and declared nothing on their form is a materially different case from one who declared the same relationship openly.

Cost: no new spend. This is a change in how existing outputs are used.

R2. Work the shortlist in tiers, and cap the intake

Divide the 500 into two review bands and work them in order: first the 138 names the model surfaced that the streaming rule missed - the cases no existing control would ever find, then the 362 corroborated by both methods, worked highest anomaly score first. Commit to a fixed number of reviews per month rather than a promise to clear the list. A queue that is worked steadily is worth more than a backlog that proves the office is overwhelmed.

Cost: existing staff time, redirected.

R3. Route the streaming queue by service line

The flagged value is concentrated in orthopedic and cardiovascular device manufacturers. Send those alerts to the reviewers who understand implant purchasing rather than into a general queue. Domain familiarity is what separates a five-minute triage from a two-hour one.

Cost: a routing rule and an agreement with two service-line leads.

6.2 Medium term (3–12 months)

R4. Set thresholds by specialty rather than one number for everyone

The results make this unavoidable. A $10,000 quarterly threshold is a meaningful signal for an internist, whose peers average $4,883 a year, and close to routine for an orthopedic surgeon, whose peers average $23,499. Publish a threshold table derived from each specialty's own distribution, for example, the 95th percentile of that specialty. Review it annually, and record the rationale. The same peer-relative logic already implemented in the pipeline supplies the numbers.

Cost: one analysis cycle plus policy committee approval. No engineering change.

R5. Put the risk calculator in front of contracting, before the contract is signed

The deployed model already accepts a specialty, an annual amount, and a concentration share and returns a tier. Wire that into the process that approves outside engagements, so a proposed arrangement can be scored before it exists rather than discovered a year later. This turns the whole system from detective to preventive, which is the single highest value change available, and it requires no new modelling.

Cost: a workflow change and access provisioning; the technical component is built.

R6. Automate the annual refresh and keep the history

CMS publishes a new program year every June. Schedule the pipeline to run on release, retain each year as a separate gold snapshot, and add year-over-year change as a feature. A clinician whose concentration jumps from 0.30 to 0.90 in twelve months is a stronger signal than either year read alone. Delta Lake's versioning makes this cheap to store and simple to compare.

Cost: scheduling configuration and incremental S3 storage.

6.3 Longer term (12+ months)

R7. Join payment exposure to internal purchasing and prescribing data

This is the recommendation that would change the nature of the answer. Open Payments can only ever establish exposure. If flagged clinicians were joined to Mephisto's own implant and formulary data, the organization could ask the question that actually matters: does a concentrated payment relationship coincide with a shift in what that clinician orders? That analysis cannot be done with public data alone and must be governed carefully since it touches patient records and clinician employment but it is where this work naturally leads.

Cost: significant. Data governance approval, a privacy review, an internal data integration, and a defined access policy.

R8. Publish the method internally before anyone is reviewed under it

Circulate the thresholds, the cluster definitions, and the tier logic to clinical leadership and let them be argued with. Three benefits follow. Clinicians who understand how the queue is built are far less likely to experience a review as an accusation. Domain experts will find edge cases the pipeline gets wrong. And a method that has already survived internal scrutiny is far easier to defend externally.

Cost: a documentation pass and two or three meetings.

6.4 What This System Should Never Be Used For

A recommendation set is incomplete without its own boundaries.

- It must not trigger automatic action. No credentialing, contracting, or disciplinary decision should follow from a tier without human review. The output is a queue position, not a finding.
- Tiers must not be shared as labels. A HIGH tier means "review this first", and it will be misread as "this person did something wrong" if it circulates outside the compliance process.
- The model must not be left to age. Payment behavior changes, the clusters will drift, and a stale model quietly stops ranking correctly. Refit and re-validate with each annual release.

7. Limitations

Stating the boundaries of the work honestly is part of making it usable.

- **Exposure, not influence.** Open Payments records what was paid. It contains nothing about what was prescribed, implanted, or ordered, so no causal claim about clinical behaviour can be made from it.
- **General payments only.** Research payments and ownership interests are published separately and are out of scope. A clinician with a large equity stake and no consulting income is invisible to this analysis.
- **A single program year.** With one year of data the system sees a level, not a trajectory. Recommendation R6 addresses this directly.
- **No ground truth.** The clustering is unsupervised because no labelled set of confirmed conflicts exists. Precision and recall cannot be computed; the 72.4% agreement between two independent methods is the strongest validation available, and it is corroboration rather than accuracy.
- **Identity matching.** Linking Mephisto's roster to CMS profile identifiers depends on name and specialty matching, which is imperfect at national scale. A production deployment would need a deliberate identity resolution step.
- **Thresholds are policy choices.** The $5,000 materiality line, the $10,000 window trigger, the 90-day window, and the 500-name shortlist are defensible but not derived from first principles. The windows are also fixed rather than sliding, so a concentration that straddles a boundary can fall below the trigger in both halves; a sliding window would close that gap at the cost of more state. They should be owned by the compliance committee, documented, and revisited.
- **Minor run-to-run variation in specialty attribution.** Where a clinician appears under more than one specialty string, the pipeline selects one without an explicit tie-break rule, so specialty-level counts can shift slightly between runs. Headline totals are unaffected. The fix is described in Section 9.3.

8. Conclusion

8.1 Summary of the Project

This project set out to answer one question for Mephisto Health System: of every clinician in the country, which few hundred should the compliance office review first, and can that answer be defended?

The answer was built as an end-to-end cloud pipeline. The CMS Open Payments file for 2025: 16,095,794 records and $2.89 billion in payments to 1,022,513 clinicians, was trimmed locally with DuckDB, landed in Amazon S3, and refined in Databricks through bronze, silver, and gold layers on Apache Spark and Delta Lake. Four workloads read the resulting gold feature table: a batch layer producing three governance indicators, a Structured Streaming job applying event-time windows and a watermark to raise concentration alerts, an unsupervised K-means model ranking peer-relative deviation, and a deployment layer that turns the trained model into a reusable scoring function and an interactive calculator. Results reach compliance staff through a four-page Power BI dashboard.

8.2 Results

- $2,894,883,382.23 in industry payments to 1,022,513 clinicians in a single program year.
- 4.29% of clinicians received more than $5,000; 12.3% were paid above their own specialty's average.
- 12,603 streaming flags covering 8,038 clinicians and $445,644,622.64, with zero records lost to the watermark.
- Four behavioural clusters at a silhouette score of 0.677, and a ranked shortlist of 500 clinicians drawn entirely from the highest-risk cluster.
- 72.4% agreement between the machine learning shortlist and the streaming rules, two methods sharing no logic.
- A deployed model: 57 specialty distributions cached, all 24 roster clinicians scored, and a live calculator validated case by case against the trained model.
- A 2,045-to-1 reduction, from more than a million clinicians to a reviewable, ranked, explainable 500.

8.3 What I Learned

The hard part was not the algorithm. K-means is a few lines of code. Deciding what to cluster on; recognizing that dollars mislead and peer-relative rank does not, took far longer than fitting the model, and it is what made the output defensible. The feature is the analysis; the algorithm is almost incidental.

Architecture is a discipline, not a diagram. The medallion layers felt like extra work in week one. By the time four workloads were reading the same gold table, the discipline was the only reason they could not disagree with each other about basic facts.

Validating between stages is not optional. Every count printed and every range checked cost minutes and repeatedly saved hours. The one place where I trusted a hard-coded assumption instead of computing it is exactly where the project's most serious defect appeared, described in Section 9.2.

Getting the model out of the notebook is its own engineering problem. Assuming deployment would be a small final step. It was not. Making a trained model usable by someone else; cached lookups, a stable function signature, validation against the original model, a path into the dashboard, took real design work, and the version that survived was not the version I first attempted.

Two mediocre detectors beat one confident one. Neither the rule nor the model is individually authoritative. Their overlap is what carries weight, and the 138 names where they disagreed turned out to be just as informative.

Explaining a system is part of building it. The information tooltips on each dashboard page, the plain-English cluster names, and the caveat repeated on every screen are not decoration. A compliance officer who cannot explain why a name is on a list will not act on the list.

9. The 3 Ws: Reflection on the Project

9.1 What Went Well

The medallion architecture paid for itself. Keeping bronze immutable meant every cleaning mistake was recoverable by rerunning rather than re-downloading eight gigabytes. Several times I changed a silver rule and simply reran; without that separation each change would have been a half-day.

Wrangling was clean, and provably so. Zero rows dropped and zero duplicates across 16 million records, with a count printed at every boundary. Splitting the packed specialty taxonomy was the highest-leverage single transformation in the project, every peer-relative feature depends on it.

The peer-relative feature design worked. Comparing clinicians to their own specialty rather than to a flat dollar line was the decision that made the results credible, and the orthopedic-versus-internal-medicine contrast in Section 4.2 demonstrates why in one line.

Two independent detectors agreed. Building a rules layer and a model layer that share no logic was more work than building either alone, and the 72.4% overlap is worth more than either result on its own.

The deployment stage landed. Despite the difficulties described below, the project ends with a working calculator rather than a notebook, and that is the difference between an analysis and a tool.

Cost stayed controlled. Trimming 90 columns to 16 locally before uploading, and separating S3 storage from Databricks compute so the cluster could be shut down between sessions, kept the whole project inside a small budget.

9.2 What Did Not Go Well

Caching in Databricks

This cost me more time than any other technical issue. Several stages recompute the same expensive aggregations, the specialty distributions in particular are needed by the KPI layer, the model, and the deployment function, and without caching, every downstream cell re-triggered the full computation over 16 million rows. Adding `.cache()` helped, but not straightforwardly: caching is lazy, so nothing is actually materialized until an action runs, and I spent time confused about why a "cached" DataFrame was still slow. Worse, cached DataFrames became stale after I changed an upstream transformation, so I was briefly reading old results and could not work out why numbers had stopped moving. Cluster memory pressure caused partial evictions and silent recomputation on top of that. The eventual approach was deliberate rather than reflexive: cache only the small, repeatedly read specialty lookup, force materialization immediately with a count, and unpersist explicitly when the stage was done.

Deploying the model in Power BI

My original plan was to have the model score clinicians inside Power BI, so a user could enter values and have the model itself respond. It kept breaking. The dependencies the scoring path needed did not resolve reliably in the Power BI environment, refreshes failed intermittently, and each failure produced an error that pointed at the environment rather than at anything I could fix in my own logic. I lost a substantial amount of time to that loop before accepting that the boundary was not going to move.

The solution was to move the computation to where the model already lived. Scoring now happens in the Databricks notebook, where the model was trained; the scored results are written back to gold as `mephisto_scored`; and Power BI simply loads that table. The calculator page presents the results and lets a reviewer explore them, but it does not attempt to run the model. This is slower to refresh. The notebook has to run before new results appear and I would not call it elegant. It is, however, stable, and it has an advantage I did not anticipate: the scoring logic now lives in exactly one place, next to the model that defines it, instead of being duplicated in a reporting tool where it could quietly drift out of agreement.

Hard-coded cluster labels: the most serious defect

K-means does not guarantee stable cluster identifiers. Cluster 2 in one run can be cluster 3 in the next, because initialization is random. I originally wrote the human-readable names and risk tiers as a dictionary keyed on those identifiers, taken from one run's output. When the model was refitted, the identifiers permuted and the labels no longer matched the clusters they described.

The visible symptom was a roster spread of 8 HIGH, 16 MEDIUM, and **zero** LOW. Statistically implausible for a roster built to contain a clear low-risk group, which is what made me look.

The fix was to stop keying on identity and start deriving names from geometry: read the fitted centroids and name each cluster from where its center actually sits on the two features. The names then follow the model rather than a remembered run, and the corrected roster spread is 8 HIGH, 8 MEDIUM, and 8 LOW. All figures in this report use the corrected mapping. I am recording this rather than quietly fixing it because it is the most instructive failure in the project: the pipeline had validation checks at every data boundary and none at all on an interpretive assumption, and that is exactly where it broke.

Re-running the stream without clearing its state

Because the streaming job writes in append mode, re-running it without deleting the output table and its checkpoint appended a second full set of flags, roughly doubling the count. The numbers did not agree with the dashboard, and it took a while to realise the pipeline was not wrong, it had simply been run twice. Output path and checkpoint have to be treated as one unit and cleared together.

Non-deterministic specialty selection

Where a clinician appears under more than one specialty string, the aggregation picks one without an explicit ordering rule. The headline totals are unaffected, but specialty-level counts shift slightly between runs, enough that I twice thought I had introduced a bug when I had not.

9.3 What I Would Do Differently Next Time

Derive every label from the data, never from a remembered run. The cluster mapping defect would not have been possible if the naming function had read the centroids from the start. The general rule I am taking from this: if a value comes from a model, compute it from the model.

Write assertions, not just print statements. I printed counts and read them. An assertion that fails loudly is better than a number that scrolls past, and would have caught both the doubled streaming output and the impossible zero-LOW tier automatically.

Design the deployment path before training the model. I treated deployment as the last step, so I discovered the Power BI limitation only after committing to it. Deciding early that scoring belongs where the model lives, and that the reporting tool only reads a table, would have saved the whole failed attempt.

Make every write idempotent. Rerunning any stage should produce the same result as running it once. Clearing checkpoint and output together, or writing with a merge rather than an append, would have removed a whole class of confusion.

Plan caching deliberately. Rather than adding `.cache()` where things felt slow, I would decide up front which small artefacts are read repeatedly, cache exactly those, force materialization, and unpersist at the end of the stage.

Keep a parameter file and a run log. The thresholds are scattered through the notebook as literals. Holding them in one place, and recording what each run produced, would make the whole thing reproducible and the thresholds easy for a policy committee to change.

Build one full slice end to end first. I built each layer completely before moving on. Taking a single specialty all the way from bronze to dashboard in the first week would have exposed both the deployment limitation and the labelling assumption while they were still cheap to fix.

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Appendix A: Glossary

Plain-English definitions of the technical terms used in this report.

Term	What it means
ACID transaction	A guarantee that a write either fully happens or does not happen at all, so a reader never sees a half-finished change.
Bronze / silver / gold	The three quality levels of the medallion architecture: raw, cleaned, and business-ready.
Checkpoint	A record of how far a streaming job has processed, so it can restart without repeating or skipping work.
Centroid	The centre point of a cluster, in the same units as the features.
Cluster (K-means)	A group of records that resemble one another on the chosen measurements.
Delta Lake	A table format that adds transactions, schema enforcement, and versioning to files in object storage.
Event time	The time an event actually happened, as opposed to the time its record arrived.
Feature	A measured property of a record that a model uses as input.
Materiality	The dollar level above which a payment is considered worth reviewing. Here, $5,000 per year.
Parquet	A column-oriented file format that compresses well and lets a query read only the columns it needs.
Percentile rank	A clinician's position within their own specialty, from 0 (lowest paid) to 1 (highest paid).
Silhouette score	A measure of clustering quality: high when clusters are tight internally and far apart from each other.
Structured Streaming	Spark's engine for processing data continuously using the same code as batch processing.
Unsupervised learning	Finding structure in data that has no labelled correct answers.
Watermark	How long a streaming job waits for late-arriving data before closing a time window.

Appendix B: Gold Feature Table Data Dictionary

The gold table holds one row per clinician and 19 columns. Schema as printed by the notebook.

Column	Type	Description
Covered_Recipient_Profile_ID	long	CMS unique clinician identifier; the grain of the table
recipient_type	string	Physician, non-physician practitioner, or teaching hospital
specialty_group	string	Top level of the CMS taxonomy string
specialty	string	Primary specialty; the peer group used throughout
state	string	Recipient's state
total_payment_usd	double	Total value received in the program year
payment_count	long	Number of individual payments

Column	Type	Description
avg_payment_usd	double	Mean payment value
max_payment_usd	double	Largest single payment
min_amount	double	Smallest single payment
distinct_manufacturers	long	Number of distinct paying companies
distinct_nature	long	Number of distinct payment types (meals, consulting, travel, …)
top_manufacturer_usd	double	Total received from the single largest paying company
top_manufacturer_share	double	Share of all money from that one company (0–1); model feature
specialty_avg_total	double	Mean per-clinician total for the specialty
specialty_std_total	double	Standard deviation of totals within the specialty
specialty_clinician_count	long	Number of clinicians in the specialty
payment_zscore_in_specialty	double	Standard deviations above or below the specialty mean
payment_pct_in_specialty	double	Percentile rank within specialty (0–1); model feature

Note. Percentile rank is retained alongside the z-score because payment distributions are heavily right-skewed — the highest-paid internal medicine clinician received $55.1 million against a specialty average of $4,883, giving a z-score of roughly 345. Percentile is the skew-robust view, and it is what feeds the model.

Appendix C: Executed Notebook Results

Key outputs as printed by the notebook, in execution order.

Stage	Printed result
Bronze load	16,095,794 rows × 16 columns; null counts recorded per column
Silver clean	0 rows dropped; 0 duplicate Record_IDs; 16,095,794 rows written
Gold write	1,022,513 rows; 19 columns; schema printed
Gold validation	0 nulls across the nine model-relevant feature columns; all range invariants hold
KPI 1	$2,894,883,382.23 across 1,022,513 clinicians
KPI 2	43,876 of 1,022,513 clinicians above $5,000 = 4.29%
KPI 3	125,919 clinicians (12%) above their own specialty average
Streaming	12,603 flags · 8,038 distinct clinicians · 131,270 payments · $445,644,622.64
Watermark check	0 records dropped as late arrivals
Silhouette sweep	k=2 0.6526 · k=3 0.6478 · k=4 0.6771 · k=5 0.6521 · k=6 0.6214 · k=7 0.6316 · k=8 0.6092
Final model	k = 4, seed = 42, silhouette 0.6771
Anomaly scoring	Maximum distance from baseline centroid = 0.7775
Shortlist	Top 500 by anomaly score; all 500 in the Concentrated High-Rank cluster
Cross-validation	362 of 500 (72.4%) also flagged by the streaming rules

Stage	Printed result
Deployment cache	57 specialty distributions cached in driver memory
Roster scoring	24 of 24 clinicians scored, 0 skipped; written to gold/mephisto_scored

Appendix D: Streaming Configuration

Setting	Value	Why
Source	Silver Delta table replayed as 24 files	Simulates arrival without changing the code
Trigger	availableNow, maxFilesPerTrigger = 1	One file per micro-batch, so incremental behaviour is observable
Window	90 days, tumbling, on event time	Turns a payment total into a rate
Watermark	30 days	Bounds state; late data beyond this is dropped
Threshold	window_total ≥ $10,000 per manufacturer	Above routine hospitality, inside real consulting range
Window type	Tumbling (fixed, non-overlapping)	Simple to explain and to audit; see the limitation in Section 7
Output mode	append	Windows are emitted once finalized
Sink	Delta table at gold/coi_stream_flags	Readable by Power BI while the job runs
Checkpoint	Dedicated S3 checkpoint location	Restart safety; must be cleared with the output on a full rerun

Validation. The job reported zero records dropped by the watermark, confirming the 30-day allowance was sufficient. The flag total of 12,603 events matches the value published to the dashboard.

Appendix E: Model Selection and Cluster Profiles

Table 8
Silhouette score by k

k	2	3	4	5	6	7	8
Silhouette	0.6526	0.6478	**0.6771**	0.6521	0.6214	0.6316	0.6092

Table 9
Fitted cluster centres and derived labels

Cluster centre (percentile, share)	Clinicians	Derived name	Tier
(0.761, 0.879)	179,484	Concentrated High-Rank	HIGH
(0.775, 0.317)	281,216	Diversified High-Rank	MEDIUM
(0.200, 0.989)	327,607	Concentrated Low-Rank	MEDIUM
(0.388, 0.520)	234,206	Baseline	LOW

Names are derived from the fitted centre positions rather than from cluster identifiers, using the rule in Appendix G. The baseline cluster is the one with the lowest sum of the two feature values, identified programmatically.

Appendix F: Deployment: Inputs, Outputs, and Validation

The deployed scoring function takes three inputs and returns a tier with its reasoning.

Input	Meaning
specialty	Must match a CMS specialty string; 57 are available in the cached population
annual_total	Total industry payments received in a year, in dollars
biggest_company	The portion of that total coming from the single largest paying company

Output field	Meaning
pct_in_specialty	Percentile position against that specialty's national distribution
top_share	biggest_company ÷ annual_total
cluster_label	Nearest cluster, named from the fitted centroids
risk_tier	HIGH, MEDIUM, or LOW
validated	Confirmation that the Python nearest-centroid result matches the fitted Spark model

Worked example. A denturist receiving $80,000 in a year, all of it from a single manufacturer, is placed at the 100th percentile of that specialty with a top-company share of 1.00, assigned to Concentrated High-Rank, returned as HIGH, and marked as matching the trained model.

Roster result. All 24 roster clinicians scored, none skipped; 8 HIGH, 8 MEDIUM, 8 LOW; written to `gold/mephisto_scored` and registered as an external Delta table for Power BI.

Appendix G: Key Code Listings

G1. Peer-relative features (Spark window functions)

```
spec = Window.partitionBy("specialty")

clinician_features = (agg
  .withColumn("specialty_avg_total", F.avg("total_payment_usd").over(spec))
  .withColumn("specialty_std_total", F.stddev("total_payment_usd").over(spec))
  .withColumn("payment_pct_in_specialty",
              F.percent_rank().over(spec.orderBy("total_payment_usd")))
  .withColumn("payment_zscore_in_specialty",
              F.when(F.col("specialty_std_total") > 0,
                     (F.col("total_payment_usd") - F.col("specialty_avg_total"))
                     / F.col("specialty_std_total")).otherwise(0.0)))
```

G2. Structured Streaming concentration flags

```
flags = (stream
    .withWatermark("event_time", "30 days")
    .groupBy(F.window("event_time", "90 days", "90 days"),
             "profile_id", "manufacturer")
    .agg(F.sum("amount").alias("window_total"),
         F.count("*").alias("payment_count"))
    .where(F.col("window_total") >= 10000.0))

q = (flags.writeStream.format("delta").outputMode("append")
```

```
    .option("checkpointLocation", checkpoint_path)
    .trigger(availableNow=True).start(flags_path))
```

G3. Anomaly score as distance from the baseline centroid

```
# the baseline cluster is identified from the data, not hard-coded
normal_id = (profile
    .withColumn("risk_sum", F.col("avg_pct") + F.col("avg_share"))
    .orderBy("risk_sum").first()["prediction"])

normal_centroid = model.clusterCenters()[normal_id].tolist()

@F.udf("double")
def dist_from_normal(features):
    import math
    return float(math.sqrt(sum(
        (f - c) ** 2 for f, c in zip(features, normal_centroid))))
```

G4. Cluster naming derived from the fitted centroids (corrected)

```
def name_cluster(pct, share):
    if pct >= 0.5 and share >= 0.7:  return "Concentrated High-Rank", "HIGH"
    if pct <  0.5 and share >= 0.7:  return "Concentrated Low-Rank",  "MEDIUM"
    if pct >= 0.5 and share <  0.7:  return "Diversified High-Rank",  "MEDIUM"
    return "Baseline", "LOW"

CLUSTER_LABELS, CLUSTER_RISK = {}, {}
for i, ctr in enumerate(model.clusterCenters()):
    CLUSTER_LABELS[i], CLUSTER_RISK[i] = name_cluster(ctr[0], ctr[1])
```

Note. This replaces a dictionary keyed on cluster identifiers taken from a single run. Because K-means identifiers permute between fits, the original mapping detached from the clusters it described when the model was refitted. See Section 9.2.

Appendix H: Additional Exhibits

Exhibit H1: original draw.io architecture export

